QUANTUMARD NATIONAL HACKATHON 2026 — PHASE 2

Smart Wall Monitoring System

IoT-Powered Structural Health Intelligence — Detect Early. Alert Instantly. Save Lives.

4+	24/7	Real-time	Cloud
Sensors	Monitoring	Alerts	Analytics

Submission Deadline: 28 February 2026, 11:59 PM IST • github.com/your-repo • Demo Video: [link]

What Is This Model?

Understanding the Smart Wall Monitoring System

A Smart, Autonomous, IoT-Based System that attaches to walls & structures — continuously sensing, analyzing, and alerting about structural health conditions without any human intervention.

📡 A Sensing Layer

Multiple sensors embedded in walls capturing crack width, vibration, moisture, and temperature simultaneously.

🧠 A Processing Brain

Arduino / NodeMCU microcontroller reads all sensor inputs, applies threshold logic, and decides when danger exists.

🚨 An Alert Engine

Instantly notifies building owners, engineers, or emergency teams via buzzer, SMS, and mobile app notifications.

Why This Model?

The Problem That Demands a Solution

5,000+

Building collapses per year

40%

Failures from undetected cracks

$250B

Annual infrastructure repair cost

72 hrs

Avg time to detect damage

Current Problem	Our Solution
✗ Manual inspection: expensive, rare, unreliable	✓ **Automated sensors: cheap, continuous, precise**
✗ Damage found only after visible cracks appear	✓ **Micro-crack detection before structural failure**
✗ No early warning — disaster strikes unannounced	✓ **Real-time SMS + buzzer alerts at first risk sign**
✗ Data not stored — no trend analysis possible	✓ **Cloud data logging enables predictive maintenance**

How This Model Works

Step-by-Step System Flow

1 📌 Sensor Attachment

Sensors fixed to wall surface at stress-critical points using adhesive mounts

2 📊 Data Capture

Continuously measures crack, moisture, vibration & temperature every 5 seconds

3 🔬 MCU Processing

Arduino reads analog/digital signals, converts and filters raw sensor data

4 ⚖️ Threshold Logic

Compares live readings against pre-configured safety thresholds in code

5 ☁️ Cloud Upload

NodeMCU sends processed data to ThingSpeak via Wi-Fi or GSM backup

6 🚨 Alert System

Buzzer triggers + SMS sent + mobile app notification on threshold breach

DECISION LOGIC

Threshold Exceeded?

YES→Alert
NO→Loop

Detailed System Workflow

Data Pipeline: Sensor → Microcontroller → Cloud → Alert

SENSORS

Crack
Moisture
Vibration
Temp

ADC/GPIO

Analog-to-
Digital
Conversion

MCU

Arduino
NodeMCU
Processing

COMM

ESP8266
Wi-Fi
GSM SIM

CLOUD

ThingSpeak
Dashboard
Analytics

OUTPUT

Buzzer
SMS
Mobile App

Continuous Feedback Loop — Data collected every 5 seconds • Cloud stores 30-day history • AI-ready for anomaly detection upgrades

Hardware Components Stack

Every component — chosen for reliability & cost-efficiency

Microcontroller

- Arduino Uno — Main processing unit
- NodeMCU ESP8266 — Wi-Fi enabled MCU

Sensors (x4 types)

- Flex/Strain Gauge — crack width
- SW-420 Vibration Sensor
- FC-28 Soil Moisture Sensor
- DHT11 Temperature Sensor

Communication

- ESP8266 Wi-Fi Module
- SIM800L GSM Module
- I2C Protocol for sensor bus

Output & Alerts

- 5V Passive Buzzer
- 16x2 LCD Display (status)
- LED Indicators (R/G)

Power Supply

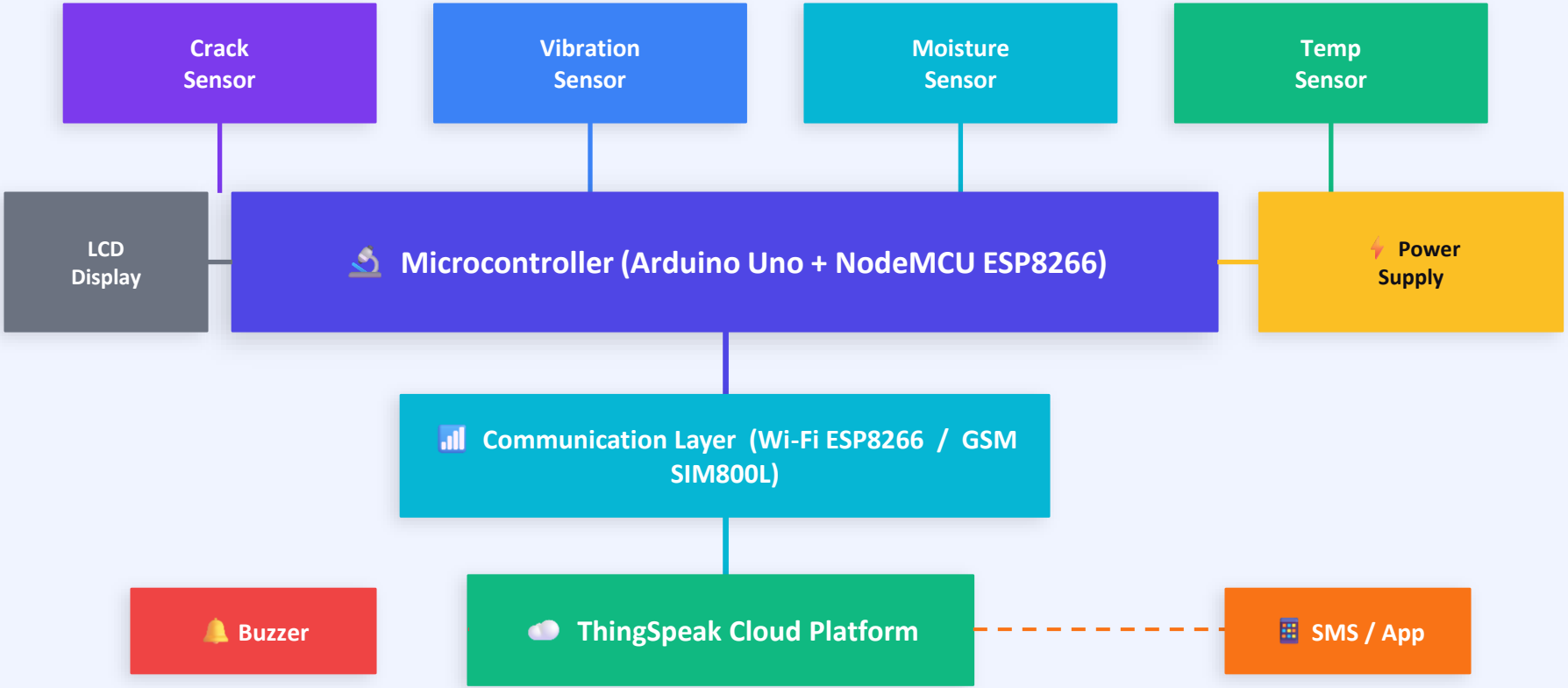
- 9V DC Adapter / Battery
- 5V Voltage Regulator (7805)
- Capacitor noise filtering

Cloud Platform

- ThingSpeak (IoT Analytics)
- MATLAB Visualisation
- Mobile App via Blynk/IFTTT

System Architecture & Block Diagram

Full hardware-to-cloud connectivity map



What Makes Us UNIQUE?

Our Competitive Differentiators vs Existing Solutions

Multi-Sensor Fusion

Combines crack + vibration + moisture + temperature in one low-cost embedded device. No competitor at this price.

Real-Time Cloud Sync

Data streams live to ThingSpeak every 5 seconds — viewable on any device globally, not just local alarms.

Sub-₹3,000 Total Cost

Enterprise SHM costs ₹5–50 lakhs. Our solution delivers 80% capability at <1% of the cost.

Zero-Infra Retrofit

Sticks onto existing walls using adhesive. No drilling, no rewiring, no structural modification needed.

AI-Ready Architecture

Threshold logic is extensible. Can plug in ML anomaly detection models for predictive failure analysis.

Dual-Comms Redundancy

Wi-Fi primary + GSM fallback ensures alerts always reach users — even when internet fails in a disaster.

How Useful Is This System?

Real-World Impact Across Sectors

85% Faster detection vs manual

24/7 Non-stop monitoring

5 sec Alert trigger latency

Applications & Use Cases

Where Smart Wall Monitoring Saves Lives

🏢 Residential & Commercial

Apartments, offices, malls — continuous crack & leak detection protecting occupants

🌉 Bridges & Dams

Critical infrastructure stress monitoring under dynamic loads & heavy traffic

🏫 Schools & Hospitals

High-occupancy public buildings where structural safety is non-negotiable

🏭 Industrial Structures

Factories with heavy vibration-generating machinery causing micro-fatigue

🏗️ Construction Monitoring

Real-time structural oversight during active construction phases

🌍 Post-Disaster Assessment

Rapid structural safety check after earthquakes or floods before re-entry

Financial Budget & Cost Analysis

Complete Bill of Materials — Prototype Build Cost

Component	Model / Spec	Qty	Unit ₹	Total ₹
Arduino Uno R3	ATmega328P, 14 I/O	1	450	450
NodeMCU ESP8266	Wi-Fi MCU, 80MHz	1	350	350
Flex / Strain Sensor	50mm flex strip	2	120	240
Vibration Sensor	SW-420 module	2	60	120
Moisture Sensor	FC-28 comparator	2	80	160
Temperature Sensor	DHT11 / LM35	2	75	150
GSM Module	SIM800L + antenna	1	420	420
Buzzer (5V)	Passive piezo	1	30	30
LCD 16x2 + I2C	Blue backlight	1	150	150
LED Indicators	Red + Green pack	1	20	20
Resistors / Caps	Assorted pack	1	50	50
Breadboard + Wires	830pt + jumper kit	1	180	180
Power Supply 9V	Adapter + 7805	1	120	120
PCB / Enclosure	Proto board + box	1	250	250

₹ 2,640

Full Prototype Cost

vs ₹5–50 Lakh

Enterprise SHM systems

99.7%

Cost Saving vs Commercial Products

Budget based on Indian market prices (2025) — subject to supplier variation ±10%

Scalability & Future Roadmap

From Prototype to Smart City Infrastructure

✅ Phase 1 — NOW

- 4 sensor nodes, single wall
- Local + cloud-based alerts
- Mobile dashboard view
- Cost: ₹2,640

🔄 Phase 2 — 6 Months

- Multi-room mesh network
- Custom PCB integration
- Android/iOS native app
- Cost: ₹15,000 / unit

🚀 Phase 3 — 1 Year

- City-level integration
- ML anomaly detection
- Govt. contracts & SaaS
- ₹499/month subscription

Future Tech: AI anomaly detection • BIM integration • Solar-powered nodes • Automated emergency dispatch

🚀 Ready to Build Safer Structures

Smart Wall Monitoring System

🎯 Problem Solved

Structural failures are preventable. Our system detects danger in real-time before it becomes disaster.

💡 Innovation

Multi-sensor IoT + cloud analytics + dual-comms at ₹2,640. Unique, scalable, patent-worthy architecture.

🌍 Impact

Protects lives across buildings, bridges & hospitals. Commercially viable at city scale via SaaS model.

📋 **PHASE 2 SUBMISSION • Deadline: 28 Feb 2026, 11:59 PM IST**

🔗 GitHub: github.com/your-repo | 🎥 Demo: youtube.com/your-video | 📄 PDF submitted via forms.gle/jCRqgbpeMVXPgzUT7